EEL 3135 Lab 2

1.1 Step 1: Template Database

```
clear all
close all
clc
%-------------------------------------
% Comparing Waveforms With Real Numbers
%-------------------------------------
f=[1:7.5:98.5];                                       % Hint: this should be a
vector or array
fs=1000;                                    % Scalar Value
dt=1/fs;                                       % Scalar Value
t=[0:dt:3];                                     % Time Vector
A=1;

for N=1:length(f)
    TemplateDataBase(N,:)=A*cos(2*pi*f(N)*t);  % Creates Database where f is
specifically indexed by N
end
```

1. Report exactly what is being stored within in 'f' and the dimesions.

'f' is the frequency of the wave, stepping 7.5Hz

2. Report exactly what is being stored in 'Fs' and the dimensions.

Frequency of sampling, which is 1000Hz described above.

3. Describe what is in "TemplateDataBase" and the dimensions.

It is an array of waves, starting at N=1, making as many waves as there are frequencies within the array of f. It then uses these frequencies to make a cosine wave with A = 1,

1.2 Step 2: Create Your Own Method to Compare with a Known Cosine Signal

```
A=1;
f_GT=8.5;
Signal_GT=A*cos(2*pi*f_GT*t);                            % Ground Truth Signal

for N=1:length(f)                                        % Iterate through the known
signal and your database
    SignalCompare(N)=sum(Signal_GT-TemplateDataBase(N,:))^2 % Store a single value
metric that provides context on the strength of the match or comparision.
    disp(SignalCompare(N))
end
```

```
SignalCompare = 
1.0000
    1.0000
SignalCompare = 1×2
    1.0000         0
     0
SignalCompare = 1×3
    1.0000         0    1.0000
    1.0000
SignalCompare = 1×4
    1.0000         0    1.0000    0.0000
   1.1148e-26
SignalCompare = 1×5
    1.0000         0    1.0000    0.0000    1.0000
    1.0000
SignalCompare = 1×6
    1.0000         0    1.0000    0.0000    1.0000    0.0000
   3.6397e-25
SignalCompare = 1×7
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000
    1.0000
SignalCompare = 1×8
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000    0.0000
   1.8213e-25
SignalCompare = 1×9
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000    0.0000 ⋯
    1.0000
SignalCompare = 1×10
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000    0.0000 ⋯
   6.2521e-25
SignalCompare = 1×11
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000    0.0000 ⋯
    1.0000
SignalCompare = 1×12
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000    0.0000 ⋯
   2.1233e-26
SignalCompare = 1×13
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000    0.0000 ⋯
    1.0000
SignalCompare = 1×14
    1.0000         0    1.0000    0.0000    1.0000    0.0000    1.0000    0.0000 ⋯
   1.6075e-26
```

1. What is the value of fGT: 8.5Hz

2. What are the dimensions of SignalGT: 1 Amp, 8.5Hz, 0 Phase Offset

3. What was the method you developed to compare the two signals? Why did you use this method. Discuss your logic in detail (More than 4 sentences). The logic doesn't have to be correct but I am looking for your thought process. Report and provide plots.

1.3 Step 3: Comparing Energy Between the Signals

```
%f_gt=??? ;
% SignalGT=A*cos(2*pi*f_gt*t);
%
% for N=1:length(f)
```

```
%     SignalEnergy(N)=??? SignalGT ??? TemplateDataBase(N,:) ??;
% end
%
% figure
% stem(f,SignalEnergy)
% grid on
% title('Comparing Signals with Real Numbers')
```

1. Using fGT = f(2) as your first experimental frequency. Provides the plots and described what happened.

2. Alter the frequency to other values within the vector f. Provides the plots and described what happened.

3. Alter the frequency to other values near the elements within the vector f but not exactly the values within the array. Start with deviations of .1 Hz, .25 Hz, .5Hz, 1 Hz, 2Hz, 3 Hz, 5Hz. Provides the plots and described what happened in great detail! (5-8 Sentences. Analyze and think about what is happening).

4. Significantly decrease and increase your time vector (1 sec, 20 seconds). Provides the plots and described what happened.

1.4 Step 4: Comparing Energy With Real and Imaginary Components

```
% f_gt=??? ;
% SignalGT=????? + ????;
%
% for N=1:length(f)
%     SignalEnergy(N)=??? SignalGT ??? TemplateDataBase(N,:) ??;
% end
%
% figure
% stem(f,SignalEnergy)
% grid on
% title('Comparing Signals using Complex Numbers')
```

1. Using fGT = f(2) as your first experimental frequency. Provides the plots and described what happened.

2. Alter the frequency to other values within the vector f. Provides the plots and described what happened.

3. Alter the frequency to other values near the elements within the vector f but not exactly the values within the array. Start with deviations of .1 Hz, .25 Hz, .5Hz, 1 Hz, 2Hz, 3 Hz, 5Hz. Provides the plots and described what happened in great detail! (5-8 Sentences. Analyze and think about what is happening)

4. Significantly decrease and increase your time vector (1 sec, 20 seconds). Provides the plots and described what happened.